When does sequence order matter? An information-theoretic taxonomy of biological representation tasks and the limits of local encoding

Zhengyi Chen
Guangxi Key Laboratory of Drug Discovery and Optimization,
Guangxi Engineering Research Center for Pharmaceutical Molecular Screening and Druggability Evaluation, School of Pharmacy, Guilin Medical University,
Guilin 541199, P. R. China
chenzhengyi@glmc.edu.cn

Ruqing Chen
GUT Geoservice Inc., Montreal, Québec, Canada
ruqing@hotmail.com

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Abstract

Biological sequence models range from amino acid composition and k-mer counters to convolutional networks and billion-parameter language models. When is a local representation sufficient, and when is global order indispensable? We answer this question by synthesizing the spectral-fibre framework into an exact information-theoretic taxonomy across the 20 431 reviewed human proteins. For any sequence S of length L over an alphabet of size q and any local order k, the total combinatorial specification cost $L \log_2 q$ splits into three non-negative telescoping terms: the composition constraint $I_{\mathrm{comp}}(S) = L \log_2 q - B_1(S)$, the resolved local-syntax information $I^{(k)}_{\mathrm{local}}(S) = B_1(S) - B_k(S)$, and the unresolved global-order entropy $I^{(k)}_{\mathrm{global}}(S) = B_k(S) = \log_2 |F_k(S)|$. Evaluating this decomposition across the human proteome for $1 \le k \le 50$ reveals a sharp separation of functional classes. At $k = 3$, 83.34% of total order entropy $B_1(S)$ is resolved on average, yet 99.64% of human proteins (20 357) retain non-trivial fibres. By $k = 5$, 99.75% of $B_1(S)$ is resolved proteome-wide. Globular catalytic enzymes ($n = 3\,449$) and receptors, channels and transporters ($n = 1\,816$) have median spectral resolution $\kappa = 5$ (95th percentile $\kappa = 7$), with only 1.04% and 0.77% requiring $k > 10$. By contrast, unresolved global order at $k > 10$ is concentrated in three architectures: C2H2 zinc-finger regulators ($n = 747$, 49.13% with $\kappa > 10$), structural tandem-repeat proteins ($n = 199$, 39.70% with $\kappa > 10$ and 9.05% with $\kappa > 25$), and hominoid segmental-duplication arrays ($n = 67$, 74.63% with $\kappa > 10$ and 29.85% with $\kappa > 25$, 95th percentile $\kappa = 257$). Combining this decomposition with our results on neural receptive fields, clinical variant copy ambiguity, zinc-finger specificity, benchmark drift ratios and symmetry quotients yields a concrete decision rule for choosing sequence encoders in computational biology.

Keywords. sequence representation, spectral fibre, fibre entropy, spectral resolution, human proteome, local encoders, protein language models, repeat proteins, representation selection.

1 Introduction

A central choice in computational biology is how to represent a biomolecular sequence or graph. Historically, two traditions have competed. The first builds fixed-length vectors from local subword or subgraph statistics: amino acid and k-mer composition, amphiphilic pseudo amino acid composition (PseAAC) [11], extended-connectivity molecular fingerprints (ECFP) [17, 19], and one-dimensional convolutional networks (CNNs) with global pooling [2, 15]. The second relies on architectures with global context—recurrent networks, state-space models [14, 13], and Transformers with positional encodings [21, 16, 12]—to capture long-range order.

Empirically, simple local models often match deep language models on bulk classification benchmarks (such as subcellular localization, broad enzyme family assignment or physicochemical property regression), yet fail catastrophically on tasks that depend on the arrangement of repeated domains or the exact copy index of a variant. Without a formal theory, this divide has remained a matter of trial and error.

In a series of companion studies [3, 4, 5, 6, 7, 8, 9, 10], we developed an exact combinatorial and algebraic theory of what local representations can and cannot see. Any encoder that depends on a sequence S only through its multiset of k-mers (and boundary $(k-1)$-mers) factors through the quotient map π_k, and is therefore constant on the *spectral fibre* $F_k(S) = \pi_k^{-1}(\pi_k(S))$. The cardinality $N_k(S) = |F_k(S)|$ of this fibre is the number of Eulerian trails of the de Bruijn multigraph $G_k(S)$, given in closed form by the BEST theorem [1, 18, 20].

This closing paper of the series synthesizes these results into a quantitative, proteome-wide taxonomy of sequence order information. Our contributions are three:

1. *Exact three-term information decomposition* (Section 2): we formulate the pointwise telescoping identity $L \log_2 q = I_{\text{comp}}(S) + I^{(k)}_{\text{local}}(S) + I^{(k)}_{\text{global}}(S)$ and evaluate it exactly across all 20 431 reviewed human proteins for 15 orders from $k = 1$ to $k = 50$ (Table 1).

2. *Functional taxonomy of order regimes* (Section 3): we partition the human proteome into six functional and structural classes and show that non-repetitive globular enzymes and membrane receptors collapse to unique sequences by $k \approx 7$–10, whereas transcriptional zinc-finger arrays, structural repeats and segmental-duplication families form a heavy tail of unresolved global order (Table 2, Figure 1).

3. *Synthesis and architectural selection guide* (Sections 4 and 5): we integrate the findings of all eight studies—from convolutional invariance theorems and ClinVar copy ambiguity to C2H2 zinc-finger reorderings, FibreBench language-model evaluations and double-stranded/molecular symmetry quotients—into a practical design rule for biological machine learning (Table 3).

2 The three-part decomposition of sequence information

Let Σ be a finite alphabet of size $q = |\Sigma|$ ($q = 20$ for standard amino acids), and let $S \in \Sigma^L$ be a sequence of length L. Specifying S among all q^L possible strings of length L requires

$$I_{\text{total}}(S) = \log_2(q^L) = L \log_2 q \quad \text{bits}$$

(4.3219 L bits for proteins).

Recall from [5] that the spectral fibre $F_k(S)$ at order $k \geq 1$ is the set of sequences $T \in \Sigma^L$ having the same k-mer multiset $\text{spec}_k(T) = \text{spec}_k(S)$ and the same prefix $T[1..k-1] = S[1..k-1]$.

Because the quotient maps refine one another as k increases, the fibres form a nested chain of subsets:

$$\Sigma^L \supset F_1(S) \supseteq F_2(S) \supseteq \cdots \supseteq F_{\kappa(S)}(S) = \{S\},$$

where $F_1(S)$ is the set of all permutations of S preserving its amino acid counts $\{c_a(S)\}_{a\in\Sigma}$, of size $N_1(S) = L!/\prod_{a\in\Sigma} c_a(S)!$, and $\kappa(S) = \min\{k : N_k(S) = 1\}$ is the *spectral resolution* of S [3, 4]. Write $B_k(S) = \log_2 N_k(S)$ for the *fibre entropy* at order k, i.e. the number of bits of sequence identity that remain unresolved once the k-spectrum of S is fixed [5].

Theorem 1 (pointwise telescoping decomposition of sequence information)**.** *For every sequence $S \in \Sigma^L$ and every order $1 \le k \le L$, the total specification cost $I_{\text{total}}(S)$ decomposes into three non-negative terms:*

$$L\log_2 q \;=\; \underbrace{\big(L\log_2 q - B_1(S)\big)}_{I_{\text{comp}}(S)} \;+\; \underbrace{\big(B_1(S) - B_k(S)\big)}_{I^{(k)}_{\text{local}}(S)} \;+\; \underbrace{B_k(S)}_{I^{(k)}_{\text{global}}(S)}. \tag{1}$$

Moreover:

1. $I_{\text{comp}}(S) = \log_2(q^L/N_1(S)) \ge 0$ *depends only on the* 1*-mer composition of S; it is the information captured by a bag-of-residues encoder ($k = 1$).*

2. $I^{(k)}_{\text{local}}(S) = \log_2(N_1(S)/N_k(S)) \ge 0$ *is monotone non-decreasing in k, equals* 0 *at $k = 1$, and equals $B_1(S)$ for all $k \ge \kappa(S)$; it is the exact order information resolved by refining from composition to k-mer statistics. Further, $I^{(k)}_{\text{local}}(S) = \sum_{j=1}^{k-1} \Delta B_j(S)$, where $\Delta B_j(S) = B_j(S) - B_{j+1}(S) \ge 0$ is the incremental information gained at scale $j + 1$.*

3. $I^{(k)}_{\text{global}}(S) = B_k(S) \ge 0$ *is monotone non-increasing in k and vanishes if and only if $k \ge \kappa(S)$; it is the exact conditional entropy $H(X \mid \pi_k(X) = \pi_k(S))$ for X uniform on $F_k(S)$, i.e. the global order information to which every order-k local encoder is provably blind [5, 6].*

Proof. Algebraic identity (1) is immediate by cancelling $B_1(S)$ and $B_k(S)$. Non-negativity and monotonicity follow from the chain of inclusions $\{S\} \subseteq F_k(S) \subseteq F_{k-1}(S) \subseteq F_1(S) \subseteq \Sigma^L$, which gives $1 \le N_k(S) \le N_{k-1}(S) \le N_1(S) \le q^L$ and hence $0 \le B_k(S) \le B_{k-1}(S) \le B_1(S) \le L\log_2 q$. Equality $B_k(S) = 0$ holds if and only if $N_k(S) = 1$, which by definition is $k \ge \kappa(S)$. □

Table 1 evaluates Theorem 1 across all 20 431 reviewed human proteins. Three quantitative facts stand out:

1. *Composition alone is a small fraction of sequence identity.* Fixing the amino acid counts ($k = 1$) accounts for a median of 0.3947 bits per residue (about 9.1% of $\log_2 20$, coming from deviations from a uniform amino acid distribution and Stirling's correction), leaving a mean of 3.8736 bits per residue in the permutation entropy $B_1(S)$.

2. *Short-range transitions ($k = 2$ to 5) absorb over* 99.7% *of bulk permutation entropy.* Dipeptides ($k = 2$) resolve 34.77% of $B_1(S)$; tripeptides ($k = 3$) resolve 83.34%; tetrapeptides ($k = 4$) resolve 98.14%; and pentapeptides ($k = 5$) resolve 99.75%. This explains why 3-mer to 5-mer composition vectors and shallow CNNs perform strongly on bulk proteome-wide benchmarks: on average, almost all permutation freedom is eliminated by $k = 5$.

3. *Bulk averages hide a persistent unresolved tail.* Even though mean B_k/B_1 is only 16.66% at $k = 3$ and 0.25% at $k = 5$, 99.64% of human proteins still have $B_3(S) > 0$, 46.52% (9 504 proteins) have $B_5(S) > 0$, 4.18% (854 proteins) have $B_{10}(S) > 0$, and 0.56% (114 proteins) have $B_{25}(S) > 0$.

order k	med. I_{comp}/L (bits/res.)	mean $I^{(k)}_{\mathrm{local}}/L$ (bits/res.)	mean B_k/L (bits/res.)	mean $I^{(k)}_{\mathrm{local}}/B_1$ (%)	mean B_k/B_1 (%)	proteins with $B_k(S) > 0$	share with $B_k(S) > 0$
1	0.3947	0.0000	3.8736	0.000%	100.000%	20 431	100.00%
2	0.3947	1.3278	2.5458	34.774%	65.226%	20 422	99.96%
3	0.3947	3.2204	0.6532	83.342%	16.658%	20 357	99.64%
4	0.3947	3.8016	0.0720	98.145%	1.855%	18 391	90.02%
5	0.3947	3.8643	0.0094	99.749%	0.251%	9 504	46.52%
6	0.3947	3.8701	0.0035	99.903%	0.097%	3 445	16.86%
8	0.3947	3.8722	0.0014	99.962%	0.038%	1 346	6.59%
10	0.3947	3.8729	0.0007	99.981%	0.019%	854	4.18%
12	0.3947	3.8732	0.0004	99.989%	0.011%	614	3.01%
16	0.3947	3.8735	0.0001	99.996%	0.004%	327	1.60%
20	0.3947	3.8735	0.0001	99.998%	0.002%	194	0.95%
25	0.3947	3.8736	0.0000	99.999%	0.001%	114	0.56%
30	0.3947	3.8736	0.0000	99.999%	0.001%	81	0.40%
40	0.3947	3.8736	0.0000	100.000%	0.000%	50	0.24%
50	0.3947	3.8736	0.0000	100.000%	0.000%	37	0.18%

Table 1: Exact decomposition of sequence information across the 20 431 reviewed human proteins of UniProt for orders $1 \le k \le 50$. Total specification cost is $\log_2 20 \approx 4.3219$ bits per residue.

functional / structural class	proteins n	med. L (aa)	mean B_3/B_1	mean B_5/B_1	$\kappa > 5$ share	$\kappa > 10$ share	$\kappa > 25$ share	med. / p95 $\kappa(S)$
Catalytic enzymes	3 449	481	17.28%	0.09%	42.22%	1.04%	0.17%	5 / 7
Receptors, channels and transporters	1 816	410	17.23%	0.08%	43.01%	0.77%	0.11%	5 / 7
Other globular and cellular proteins	14 153	380	16.23%	0.19%	45.00%	2.18%	0.35%	5 / 8
C2H2 zinc-finger regulators	747	560	17.80%	1.40%	88.49%	49.13%	2.54%	10 / 23
Structural tandem-repeat proteins	199	195	25.80%	4.14%	86.43%	39.70%	9.05%	10 / 41
Segmental-duplication arrays	67	638	19.99%	1.70%	97.01%	74.63%	29.85%	16 / 257

Table 2: Information decomposition and spectral resolution $\kappa(S)$ across six classes of the human proteome. The upper three classes follow the random-sequence law $\kappa \sim 2\log_{20} L + O(1)$ (95th percentile $\kappa \le 8$); the lower three classes form the repeat-driven tail.

3 A proteome-wide taxonomy of order regimes

To see which biological functions sit in which order regime, we partitioned the 20 431 reviewed human proteins into six non-overlapping classes based on UniProt family and functional annotations (Table 2 and Figure 1).

Table 2 and Figure 1 establish three natural regimes of sequence encoding:

Regime I: Composition-dominated and short-motif tasks ($k = 1$ to 5). Bulk physicochemical properties (molecular weight, isoelectric point, average hydropathy, overall charge) are exact functions of the 1-mer spectrum ($k = 1$). Short linear targeting motifs and cleavage sites reside at $k = 3$ to 5. At $k = 3$, however, every class still retains 16–26% of its order entropy $B_1(S)$, and a single protein of 300 residues can have 2^{239} Eulerian reconstructions [4]. Thus $k = 3$ separates reference proteins empirically while leaving individual sequences heavily under-determined.

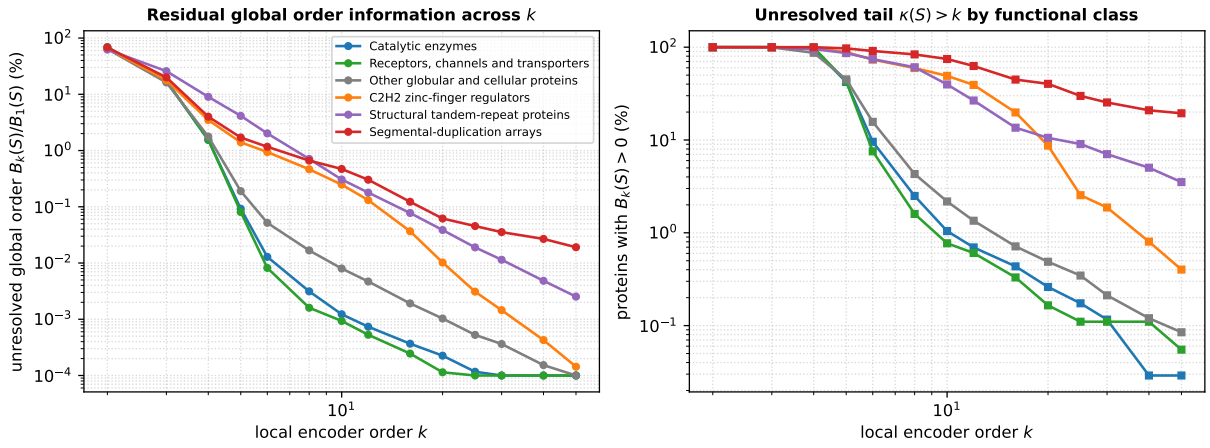


Figure 1: Left: mean unresolved global order $B_k(S)/B_1(S)$ (%) as a function of local encoder order k across the six protein classes. Right: percentage of proteins in each class whose sequence remains unresolved at order k ($B_k(S) > 0$, i.e. $\kappa(S) > k$).

Regime II: Globular enzymes and membrane receptors ($k = 5$ to 10). For catalytic enzymes ($n = 3\,449$), membrane receptors, ion channels and transporters ($n = 1\,816$), and general globular proteins ($n = 14\,153$), the residual order entropy B_5/B_1 drops below 0.2% at $k = 5$. Their median spectral resolution is $\kappa = 5$, and 95% of all enzymes and receptors are uniquely reconstructed from their 7-mer spectrum (p95 $\kappa = 7$; over 98.9% by $k = 10$). Because globular folds avoid long exact repeats to prevent misfolding and aggregation, their sequences obey the random-sequence scaling $\kappa(S) \approx 2 \log_{20} L$. For phenotypes on these classes, a local receptive field of $R \geq 10$ is injective on over 98.9% of wild-type proteins.

Regime III: Repeat-array and long-range order tasks ($k > 10$ to $k > 25$). The three lower classes in Table 2 violate random-sequence scaling by up to two orders of magnitude:

- *C2H2 zinc-finger regulators ($n = 747$, including KRAB-ZFPs):* 88.49% require $k > 5$, 49.13% require $k > 10$, and 2.54% (19 proteins in this broader class; 17 with annotated canonical C2H2 domains in [8]) require $k > 25$. Their median κ is 10 and their 95th percentile is 23, driven by conserved linker-helix repeats and recent tandem finger duplications.
- *Structural tandem-repeat proteins ($n = 199$, mucins, collagens, keratin-associated proteins, filaggrin):* mean B_3/B_1 is 25.80%, 39.70% require $k > 10$, 9.05% require $k > 25$, and p95 $\kappa = 41$.
- *Hominoid segmental-duplication arrays ($n = 67$, NBPF, GOLGA6, NPIP, POTE):* 74.63% require $k > 10$, 29.85% require $k > 25$, median $\kappa = 16$, and p95 $\kappa = 257$ (reaching $\kappa = 560$ in NBPF20 [4]).

4 Synthesis across the eight-part series

Table 3 summarizes how the seven companion studies and the present work fit together as a single mathematical and empirical programme.

Together, the eight studies establish four cross-cutting principles:

part	focus	core mathematical object	principal empirical finding
I [3, 4]	Proteome spectral atlas & Separation ≠ Resolution	De Bruijn multigraph $G_k(S)$, BEST count $N_k(S)$, resolution $\kappa(S)$	100% of unique human proteins separated at $k = 3$, yet 99.64% unresolved (854 at $k = 10$, 114 at $k = 25$)
II [5]	Quotient geometry & information loss	Quotient lattice $(\mathcal{L}, \preceq)$, fibre entropy $B_k(S)$, diameter $D_k(S)$	Exact mutual-information ceiling $H(S) - \mathbb{E}[B_k]$ and Fano error lower bound; $D_k(S) = \Theta(L)$
III [6]	Neural architectural boundaries	Window integral $\int F \, d\mu_S^R$; symmetric vs. sign-aware relative bias; SSM tail bound	15 PyTorch architectures tested in float64 (2 700 pairs): zero violations; attention pooling & symmetric clip stay in F_R
IV [7]	Genomic reachability & ClinVar variants	Transition operators $\mathcal{M}_{\text{genomic}}$, reachable fibre F_k^{bio}, copy twins	Single variants leave F_k; 36.5% copy ambiguity in NBPF at $k = 10$; strong depletion in ClinVar (4 P/LP twins at $k = 10$, 0 at $k = 25$)
V [8]	Phenotypic insufficiency (C2H2 zinc fingers)	Finger-multiset conservation ($k \geq 24$), block swap across $(k-1)$-mers	883/1 178 (75%) $k = 25$ mates across 17 C2H2 proteins reorder recognition helices (17/31 in ZNF91); KLF18 negative control
VI [9]	FibreBench evaluation of protein language models	Same-protein substitution yardstick, drift ratio $\rho = \Delta_{\text{fibre}}/\Delta_{\text{sub}}$	Across 258 pairs (1 016 sequences), ESM-2 (8M–650M) median $\rho = 0.49$–0.81 at $k = 25$ (70% sub-mutational)
VII [10]	Symmetry-constrained fibres (DNA & ECFP)	$\mathbb{Z}/2$ (-1)-eigenlattice Λ^- on double cover; coarsened symmetric window Ω_r	ϕX174 $\lvert\text{DS}(S, 10)\rvert = 3.19 \cdot 10^{13}$ vs. $1.37 \cdot 10^{36}$ double-cover circuits; exact RDKit ECFP4/6 collision pairs
VIII	Information decomposition & selection guide	Telescoping identity $L \log_2 q = I_{\text{comp}} + I_{\text{local}}^{(k)} + B_k$	Globular enzymes/receptors resolve by $\kappa \leq 7$ (p95); repeat families form the $k > 10$ tail; four-step practitioner guide

Table 3: Overview of the eight-paper programme on the spectral fibres of biological sequence and molecular encoders.

1. *Empirical separation on a dataset does not imply injectivity on sequence space* (Parts I–II). A representation can assign distinct vectors to all 20 375 distinct human reference sequences at $k = 3$ while collapsing 2^{100} or more alternative Eulerian arrangements onto each sequence's vector.

2. *Architectural complexity does not break locality without asymmetric positional input* (Part III). Adding depth, residual connections, gating, attention pooling, unpositioned self-attention or symmetrically clipped relative biases only changes the effective window R; escaping F_R requires absolute positions, strides, sign-aware/unclipped relative biases, or non-decaying state spaces.

3. *Even escaping architectures suffer from pooling attenuation on long repeats* (Part VI). Although ESM-2 (RoPE) and ProtBERT (absolute position) lie outside the invariant class in principle, global mean pooling averages out phase shifts across transposed blocks, leaving 70% of $k = 25$ fibre rearrangements smaller in embedding shift than a single amino acid substitution.

4. *Where local models are blind, short-read clinical databases are also nearly blind* (Parts IV–V).

Order-25 fibre rearrangements dramatically alter predicted DNA-binding motifs in KRAB zinc-finger proteins (Part V), and copy-ambiguous twin positions cover 23–37% of mucins and NBPF proteins (Part IV). Yet in ClinVar, copy-ambiguous pathogenic missense variants are depleted to 4 at $k = 10$ and 0 at $k = 25$, because short-read sequencing discards multi-mapping reads in identical repeats.

5 A practitioner's decision rule for sequence representation

For a given biological prediction problem with target phenotype $Y\colon \mathcal{X} \to \mathcal{Y}$ on a protein or nucleic-acid class $\mathcal{X}$, our results reduce encoder selection to four verifiable checks:

1. **Check the spectral resolution $\kappa(\mathcal{X})$ of the target sequences.** Compute $\kappa(S)$ (in $O(L \log L)$ time via suffix arrays or de Bruijn Eulerian degree checks) for the sequences in $\mathcal{X}$. If the target class consists of globular enzymes, transporters or non-repetitive cellular proteins, 95% of sequences satisfy $\kappa(S) \leq 7$ and 99% satisfy $\kappa(S) \leq 10$ (Table 2). Any encoder with receptive field $R \geq 10$ is injective on almost the entire domain.

2. **Check phenotypic sufficiency on the fibre $F_R(S)$.** If $\mathcal{X}$ contains repeat proteins with $\kappa(S) > R$, test whether the target phenotype Y is invariant under elementary fibre moves (exchanging segments bounded by repeated $(R-1)$-mers, or swapping homologous variants between identical repeat copies). For bulk stability or composition-driven properties, Y is approximately fibre-invariant and local encoders suffice. For order-sensitive phenotypes—such as tandem C2H2 zinc-finger DNA specificity (Part V) or domain-specific pathogenic mutations in repeat arrays (Part IV)—any encoder confined to F_R incurs an irreducible error of at least $\frac{1}{2}\, d(Y(S), Y(T))$ (Part V, Theorem 4).

3. **Choose an escaping mechanism that matches the read-out.** When $R < \kappa(S)$ and Y is order-sensitive, do not rely on deeper stride-1 convolutions, attention pooling or symmetrically clipped relative biases (Part III, Table 1). Include absolute positional encodings, sign-aware or unclipped relative encodings (Shaw, T5, RoPE, ALiBi), or long-memory state-space layers ($\rho \to 1$).

4. **Avoid global mean pooling when domain order or copy index is the signal.** As FibreBench demonstrates (Part VI), even a 650M-parameter RoPE Transformer loses most of its sensitivity to block transpositions when its residue states are collapsed by global mean pooling (median $\rho = 0.81 < 1$). Use position-retaining read-outs (per-residue heads, cross-attention to positional queries, or flattened/strided read-outs) whenever the downstream task depends on which repeat copy is modified or how repeat domains are ordered.

Data and code availability

The script `code/order_taxonomy.py` computes the exact information decomposition (Table 1), the functional class breakdown (Table 2) and Figure 1 from the proteome fibre catalogue, writing `results/order_taxonomy_output.txt` and `results/order_taxonomy_summary.json`. Code, results and manuscript sources are available at `https://github.com/Ruqing1963/order-taxonomy`.

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